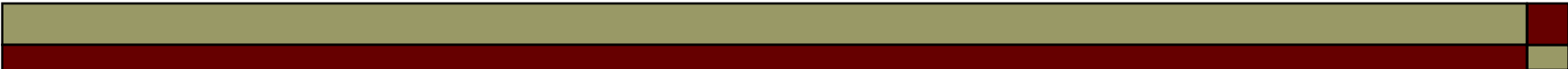


Open Development: Is the “Open Source” Analogy Relevant to Biotechnology?

Alan G. Isaac & Walter G. Park

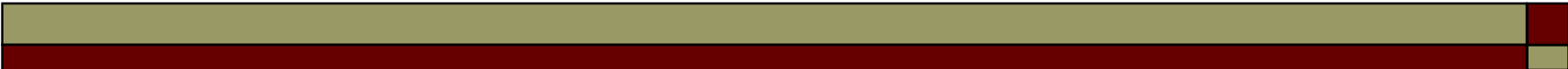
Issues:

1. Open-Biotech: Background/Examples
2. Implications for Innovation
3. Implications for Developing Countries



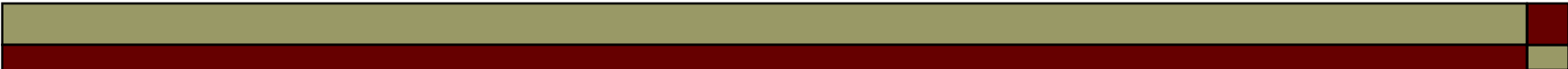
1. Varieties of Openness

- Open Standards
 - Technological standards (data exchange format, annotation systems, repository standards, etc.)
- Open Innovation
 - Share findings, participatory research
- Open Licensing
 - Non-exclusive, royalty-free
 - Share improvements/modifications under same terms
 - Does not preclude commercialization & patenting



Examples

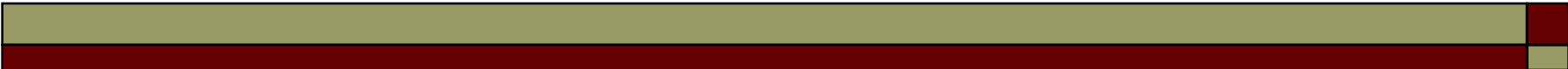
- CAMBIA – BIOS – BioForge
- SNP Consortium
- International Hap Map Project
- Tropical Disease Initiative



2. Implications for Innovation

A. Underlying Principles:

- Free-Revealing
 - i.e. it may pay to reveal secrets
- Collective Invention
 - Sharing & experimentation, under technological uncertainty
 - Profiting from complementary assets
- User Innovation Theory
 - End-users with specific needs play active role
 - E.g. Patient involvement in R&D



B. Concerns:

- Incentives to Commercialize
- Crowding out of Proprietary Innovation
- Capital Intensity of Biotechnology
- Regulatory Approval
- Comparative Disadvantages



3. Implications for Developing countries

A. Adaptation & Learning:

- Is Open Development a suitable mode of innovation?
- Can Open Development contribute to human capital accumulation?

B. Concerns:

- Is Open-Biotech a viable business model?
- Role in developing country diseases?